

The Bloch Sphere and Single-Qubit Gates

29 September, 1 October • Objective 3

Two complex amplitudes, four real parameters, minus normalization, minus global phase, leaves two — and two parameters is a sphere. The Bloch sphere is not a visualization aid bolted on afterward; it is the exact state space of one qubit, and every single-qubit gate is a rotation of it.

Constructing the sphere

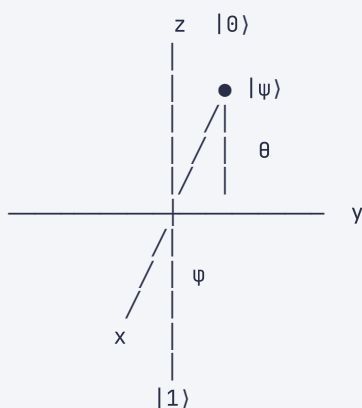
Start from $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ and remove what is not physical. Write $\alpha = r_0 e^{i\gamma_0}$, $\beta = r_1 e^{i\gamma_1}$. Global phase is unobservable, so factor out $e^{i\gamma_0}$:

$$|\psi\rangle \approx r_0|0\rangle + r_1 e^{i(\gamma_1 - \gamma_0)}|1\rangle$$

Only the **relative** phase $\phi = \gamma_1 - \gamma_0$ survives. Normalization $r_0^2 + r_1^2 = 1$ lets you write $r_0 = \cos(\theta/2)$, $r_1 = \sin(\theta/2)$. The general state is:

$$|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\phi} \sin(\theta/2)|1\rangle$$

$$\theta \in [0, \pi] \quad \phi \in [0, 2\pi)$$



The **half-angle** is the point people trip over: $\theta = \pi$ maps $|0\rangle$ to $|1\rangle$, so states orthogonal in Hilbert space are **antipodal** on the sphere, not perpendicular. $|0\rangle$ and $|1\rangle$ are opposite poles despite $\langle 0|1\rangle = 0$.

The six cardinal points:

AXIS	STATE	θ, ϕ
+z	$ 0\rangle$	0, —
-z	$ 1\rangle$	π , —
+x	$ +\rangle = (0\rangle + 1\rangle)/\sqrt{2}$	$\pi/2$, 0
-x	$ -\rangle = (0\rangle - 1\rangle)/\sqrt{2}$	$\pi/2$, π
+y	$ +i\rangle = (0\rangle + i 1\rangle)/\sqrt{2}$	$\pi/2$, $\pi/2$
-y	$ -i\rangle = (0\rangle - i 1\rangle)/\sqrt{2}$	$\pi/2$, $3\pi/2$

The Bloch vector components are the Pauli expectation values — a fact that makes the picture operationally concrete:

$$\mathbf{r} = (\langle X \rangle, \langle Y \rangle, \langle Z \rangle) \quad |\mathbf{r}| = 1 \text{ for pure states}$$

Analogy boundary — the sphere as a state-space diagram. Reading it as a phase-space or state-machine diagram works well: points are states, gates are transitions. Two boundaries. First, only **one qubit** gets a sphere; two qubits need 6 real parameters and no three-dimensional picture exists — entanglement is invisible here, which is why the Bloch sphere quietly stops helping in Week 8. Second, points *inside* the ball are **mixed states**, with $|\mathbf{r}| < 1$ measuring purity. The center is the maximally mixed state $I/2$, and decoherence (Week 13) is the state vector shrinking toward it.

Pauli gates

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Each is a π rotation about its axis.

X — bit flip, the quantum NOT. $X|0\rangle = |1\rangle$, $X|1\rangle = |0\rangle$. Rotation by π about $\hat{x}$. Note it leaves $|+\rangle$ and $|-\rangle$ fixed (they lie on the rotation axis).

Z — phase flip. $Z|0\rangle = |0\rangle$, $Z|1\rangle = -|1\rangle$. Rotation by π about $\hat{z}$. It leaves $|0\rangle$ and $|1\rangle$ alone up to phase, and swaps $|+\rangle \leftrightarrow |-\rangle$. **Z has no classical analogue** — it does nothing observable in the computational basis, yet it is essential, because interference converts phase into amplitude.

Y — both, with a phase: $Y = iXZ$.

All satisfy $P^2 = I$: applying twice returns the identity, as a π rotation twice is a 2π rotation.

Hadamard

$$H = (1/\sqrt{2}) \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad \begin{array}{l} H|0\rangle = |+\rangle \\ H|1\rangle = |-\rangle \end{array} \quad \begin{array}{l} H|+\rangle = |0\rangle \\ H|-\rangle = |1\rangle \end{array}$$

H is a π rotation about the diagonal axis $(\hat{x}+\hat{z})/\sqrt{2}$, which exchanges the x and z axes. It maps the computational basis to the superposition basis and back, and $H^2 = I$.

This is the most-used gate in the course: it is how you *create* superposition from a classical input, and — applied again at the end — how you convert phase differences back into measurable amplitude differences. That sandwich, $H \rightarrow$ phase manipulation $\rightarrow H$, is the skeleton of Deutsch's algorithm (Week 11) and of Grover's diffusion operator (Week 13).

Phase gates

$$S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} = \sqrt{Z} \quad T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix} = \sqrt{S}$$

S rotates $\pi/2$ about $\hat{z}$; T rotates $\pi/4$. Both act only on the $|1\rangle$ component's phase.

T matters disproportionately. The Clifford group — generated by H, S, and CNOT — is **efficiently simulable classically** (Gottesman–Knill theorem). Adding T makes the set universal. So T gates are the source of quantum advantage, and in fault-tolerant architectures they are also by far the most expensive to implement, requiring magic-state distillation. "T-count" is a primary cost metric in quantum compilation — the number to watch when you read resource estimates in Week 16.

Rotation gates

For continuous control, the parameterized rotations:

$$\begin{aligned} R_x(\theta) &= e^{-i\theta X/2} = \begin{bmatrix} \cos(\theta/2) & -i \sin(\theta/2) \\ -i \sin(\theta/2) & \cos(\theta/2) \end{bmatrix} \\ R_y(\theta) &= \begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix} & R_z(\theta) &= \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix} \end{aligned}$$

The half-angles again: $R_x(2\pi) = -I$, not I . A qubit requires a 4π rotation to return exactly — spinor behavior, and a real physical effect confirmed by neutron interferometry.

Any single-qubit unitary decomposes as three rotations (up to global phase):

$$U = R_z(\gamma) R_y(\beta) R_z(\alpha)$$

This is what a transpiler does when it maps your circuit onto a device's native gate set.

Try it yourself. In `bloch_sphere.html`, verify these by hand:

1. **H** then **Z** then **H** — equals **X**. Watch the axis exchange do the work.
2. **T** eight times — returns to start ($T^8 = I$), since $8 \times \pi/4 = 2\pi$.
3. **X** on $|+\rangle$ — nothing moves. The state is on the rotation axis.
4. **S** then **S** — equals **Z**. Confirms $S = \sqrt{Z}$.

Key takeaways

- One qubit's state space is exactly the Bloch sphere surface: $|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\phi}\sin(\theta/2)|1\rangle$, after removing normalization and global phase.
- Half-angles mean orthogonal states are antipodal; the Bloch vector is $(\langle X \rangle, \langle Y \rangle, \langle Z \rangle)$.
- Pauli gates are π rotations about their axes; **Z** changes only phase and has no classical counterpart.
- **H** exchanges the x and z axes, creating superposition and converting phase back to amplitude — the core of every algorithm in this course.
- Clifford gates (**H**, **S**, **CNOT**) are classically simulable; **T** supplies universality and dominates fault-tolerant cost.

Self-check

Q1 (Objective 3) — **Z** applied to $|0\rangle$ or $|1\rangle$ produces no measurable change in the computational basis. Explain why **Z** is nonetheless indispensable, and give the minimal circuit that makes its effect observable.

Q2 (Objective 3) — The Gottesman–Knill theorem says circuits of H, S, and CNOT are classically simulable in polynomial time, despite producing entanglement and superposition. State what this implies about which resource is actually responsible for quantum advantage.

Check your answers

A1 — Z is a **relative-phase** operation, and relative phase is invisible to measurement in the basis it acts on but fully visible after a basis change. Measuring $Z|\psi\rangle$ in the computational basis gives outcome probabilities $|\alpha|^2$ and $|\beta|^2$, identical to $|\psi\rangle$ — the sign on β vanishes under the modulus.

The minimal circuit that exposes it is **H–Z–H**, which equals X:

$ 0\rangle \xrightarrow{H} \xrightarrow{Z} \xrightarrow{H} = 1\rangle$	measured: 1 with certainty
$ 0\rangle \xrightarrow{H} \xrightarrow{\quad\quad\quad} \xrightarrow{H} = 0\rangle$	measured: 0 with certainty

Tracing it: H sends $|0\rangle \rightarrow (|0\rangle + |1\rangle)/\sqrt{2}$. Z flips the sign of the $|1\rangle$ component, giving $(|0\rangle - |1\rangle)/\sqrt{2} = |-\rangle$. The final H maps $|-\rangle \rightarrow |1\rangle$. A phase that was unmeasurable in the middle of the circuit becomes a deterministic bit flip at the end.

This is the mechanism behind every algorithm in the course. Computation happens in phases; Hadamards convert phase into amplitude so it can be read. Deutsch's algorithm is exactly this pattern with an oracle in place of Z, and **phase kickback** is the name for the trick.

A2 — It implies that **entanglement and superposition are not sufficient** for quantum advantage. Clifford circuits generate both — CNOT on $|+\rangle|0\rangle$ produces a maximally entangled Bell state — yet the stabilizer formalism tracks the whole evolution with polynomial classical resources. A machine running only Clifford gates, however many qubits, offers no speedup.

What supplies the advantage is **non-stabilizer resource**, usually called *magic*, contributed by gates like T that rotate outside the Clifford group. The practical consequence is that T-count, not qubit count, is the meaningful measure of a fault-tolerant algorithm's cost — and in surface-code architectures T gates require magic-state distillation, consuming many physical qubits and cycles per logical T.

The point to carry into Week 16: "how many qubits" is close to the wrong question. A thousand-qubit Clifford-only device is classically simulable. The figures that matter are logical qubit count, gate fidelity, and T-count — which is why current NISQ hardware, measured in physical qubits, sits much further from useful cryptanalysis than headline numbers suggest.